RSpec. Codereview

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Как избавить себя и команду от боли при работе с RSpec?

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ruby
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Codereview

План

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3. Тестовые классы
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Конфигурация

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helper конфиг 1/2

- spec_helper
- fast_spec_helper
- rake_helper
- rubocop_spec_helper

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec

helper конфиг 2/2

```
RSpec.configure do |config|
  config.define_derived_metadata(file_path: %r{spec/rubocop}) do |metadata|
    metadata[:type] = :rubocop
  end

  config.include RuboCop::RSpec::ExpectOffense, type: :rubocop
  config.include_context 'config', type: :rubocop
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/rubocop_spec_helper.rb

Tags

```
it 'should tell about pipeline mode', :postgresql_14 do
  @conn.enter_pipeline_mode
  expect(@conn.inspect).to match(/ pipeline_status=PQ_PIPELINE_ON/)
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/connection_spec.rb

```
# TODO: each spec file should require its dependencies.
# Maybe we'll get there one day
require "rom"
require "rom/sql"
require "rom/memory"
require "rom/repository"
require "rom/changeset"
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/spec_helper.rb

Определение группы

```
config.define_derived_metadata file_path: %r{/suite/} do |metadata|
  metadata[:group] = metadata[:file_path]
                     .split('/')
                     .then { |parts| parts[parts.index('suite') + 1] }
                     .to_sym
end

%i[rom compat].each do |group|
  config.when_first_matching_example_defined group: group do
    require_relative "support/#{group}"
  end
end
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/spec_helper.rb

Helper 1/3

```
RSpec.describe ROM::SQL::Associations::ManyToMany, helpers: true do
```

https://github.com/rom-rb/rom-sql/blob/beb115/spec/integration/associations/many_to_many_spec.rb

Helper 2/3

```
config.include(Helpers, helpers: true)
```

https://github.com/rom-rb/rom-sql/blob/beb115/spec/spec_helper.rb

Helper 3/3

```
Dir[root.join("shared/**/*.rb")].sort.each { |f| require f }
Dir[root.join("support/**/*.rb")].sort.each { |f| require f }
```

https://github.com/rom-rb/rom-sql/blob/beb115/spec/spec_helper.rb

Sandboxed

```
RSpec.configure do |c|
  c.around do |ex|
    RSpec::Core::Sandbox.sandboxed do |config|
      # If there is an example-within-an-example,
      # we want to make sure the inner example
      # does not get a reference to the outer example
      # (the real spec) if it calls
      # something like `pending`
      config.before(:context) { RSpec.current_example = nil }

      config.color_mode = :off

      orig_load_path = $LOAD_PATH.dup
      ex.run
      $LOAD_PATH.replace(orig_load_path)
    end
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/support/sandboxing.rb

```ruby
require "warning"

Warning.ignore(%r{rspec/core})
Warning.ignore(%r{rspec/mocks})
Warning.ignore(/codacy/)
Warning[:experimental] = false if Warning.respond_to?(:[])
```

https://github.com/dry-rb/dry-monads/blob/704c1b/spec/support/warnings.rb

Контекст

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```
▾ fixtures/
  ▾ app/persistence/
    ▸ commands/
    ▸ mappers/
    ▸ my_commands/
    ▸ my_mappers/
    ▸ my_relations/
    ▸ relations/
  ▾ auto_loading/
    ▾ app/
      ▸ commands/users/
      ▸ mappers/
      ▸ relations/
    ▸ persistence/
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/fixtures

```ruby
RSpec.shared_context 'changeset / database' do
  include_context 'changeset / database setup'

  before do
    %i[tags tasks books posts_labels posts users labels
       reactions messages].each { |table| conn.drop_table?(table) }

    conn.create_table :users do
      primary_key :id
      column :name, String
    end

    conn.create_table :books do
      primary_key :id
      foreign_key :author_id, :users, on_delete: :cascade
      column :title, String
      column :created_at, Time
      column :updated_at, Time
    end
  end
end
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/shared/rom/changeset/database.rb

attr_reader

```
context 'when used in combination with the BisectDRbFormatter', :slow do
  include FormatterSupport

  attr_reader :server # <<<<<<<<<

  around do |ex|
    Bisect::Server.run do |the_server|
      @server = the_server
      ex.run
    end
  end

  def run_formatter_specs
    RSpec.configuration.drb_port = server.drb_port
    run_rspec_with_formatter('bisect-drb')
  end

  it 'receives suite results' do
    results = server.capture_run_results(['spec/rspec/core/resources/formatter_specs.rb']) do
    end
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/rspec/core/bisect/server_spec.rb

Перменные вне блоков

```
RSpec.describe(Dry::Monads) do
  let(:m) { described_class }
  list = Dry::Monads::List # <<<<<<<<<<<<

  it "builds a list with List[]" do
    expect(instance.make_list).to eql(list[1, 2, 3])
  end
end
```

https://github.com/dry-rb/dry-monads/blob/704c1b/spec/integration/monads_spec.rb

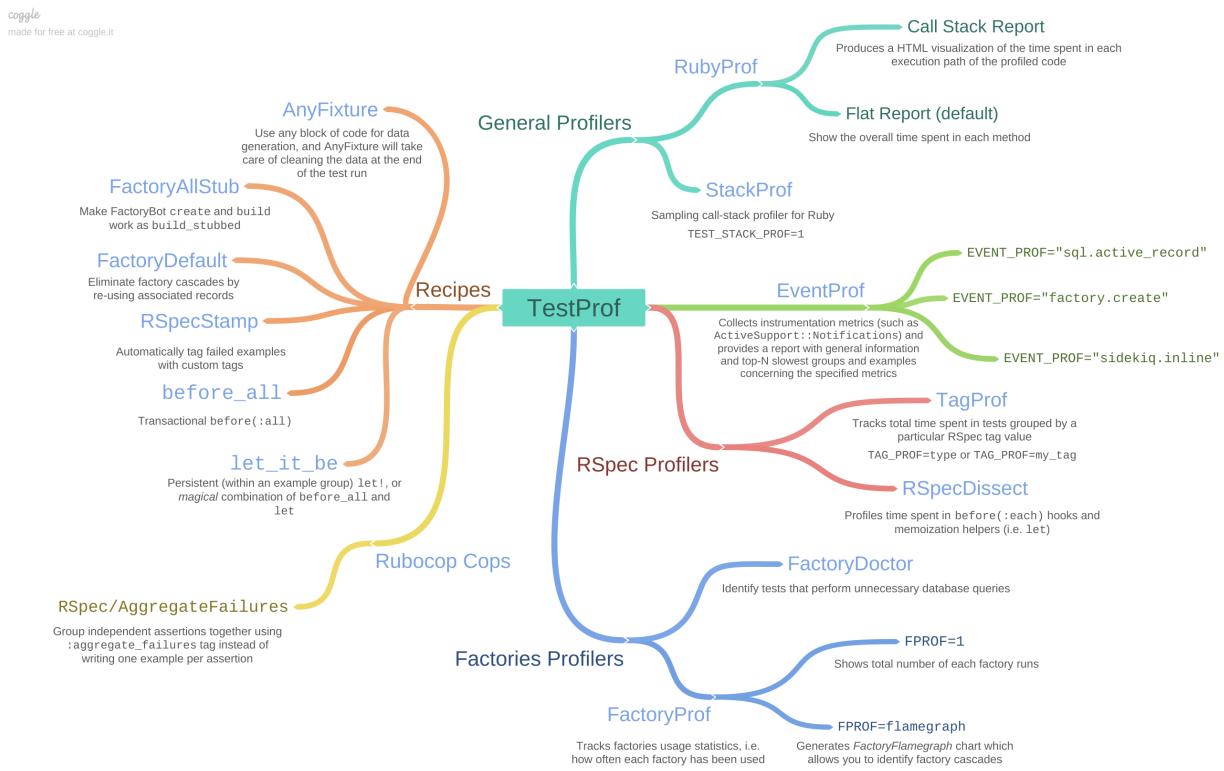
let_it_be

```
RSpec.describe Projects::BranchesByModeService, feature_category: :source_code_management do
  let_it_be(:user) { create(:user) }
  let_it_be(:project) { create(:project, :repository) }

  let(:finder) { described_class.new(project, params) }
  let(:params) { { mode: 'all' } }
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/services/projects/branches_by_mode_service_spec.rb

TestProf


coggle
made for free at coggle.it
TestProf
Recipes
AnyFixture
Use any block of code for data generation, and AnyFixture will take care of cleaning the data at the end of the test run
FactoryAllStub
Make FactoryBot create and build work as build_stubbed
FactoryDefault
Eliminate factory cascades by re-using associated records
RSpecStamp
Automatically tag failed examples with custom tags
before_all
Transactional before(:all)
let_it_be
Persistent (within an example group) let!, or magical combination of before_all and let
Rubocop Cops
RSpec/AggregateFailures
Group independent assertions together using :aggregate_failures tag instead of writing one example per assertion
General Profilers
RubyProf
Call Stack Report
Produces a HTML visualization of the time spent in each execution path of the profiled code
Flat Report (default)
Show the overall time spent in each method
StackProf
Sampling call-stack profiler for Ruby
TEST_STACK_PROF=1
EventProf
Collects instrumentation metrics (such as ActiveSupport::Notifications) and provides a report with general information and top-N slowest groups and examples concerning the specified metrics
EVENT_PROF="sql.active_record"
EVENT_PROF="factory.create"
EVENT_PROF="sidekiq.inline"
RSpec Profilers
TagProf
Tracks total time spent in tests grouped by a particular RSpec tag value
TAG_PROF=type or TAG_PROF=my_tag
RSpecDissect
Profiles time spent in before(:each) hooks and memoization helpers (i.e. let)
Factories Profilers
FactoryDoctor
Identify tests that perform unnecessary database queries
FactoryProf
Tracks factories usage statistics, i.e. how often each factory has been used
FPROF=1
Shows total number of each factory runs
FPROF=flamegraph
Generates FactoryFlamegraph chart which allows you to identify factory cascades

`as_null_object` 1/2

```
allow(config).to receive(:logger).and_return(double.as_null_object)
```

https://github.com/vcr/vcr/blob/8f5636/spec/lib/vcr/cassette/http_interaction_list_spec.rb

`as_null_object` 2/2

```
logger = double.as_null_object

expect { logger.info }.not_to raise_error
expect(logger.foo.bar).to be_nil?
```

Инкапсуляция

```
describe 'clean_outdated_http_interactions' do
  before(:each) do
    subject.instance_variable_set(:@clean_outdated_http_interactions, true)
    subject.instance_variable_set(:@previously_recorded_interactions,
                                  subject.instance_variable_get(:@new_recorded_interactions))
    subject.instance_variable_set(:@new_recorded_interactions, [])
  end
end
```

https://github.com/vcr/vcr/blob/8f5636/spec/lib/vcr/cassette_spec.rb

Тестовые классы

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Анонимные классы 1/2

```
subject do
  Class.new do
    def zero
      nil
    end

    # ...
  end.new
end
```

https://github.com/ruby-concurrency/concurrent-ruby/blob/1982b9/spec/concurrent/async_spec.rb

Анонимные классы 2/2

```
it "shouldn't accept invalid return from fit_to_copy_get" do
  tm = Class.new(PG::TypeMapInRuby) do
    def fit_to_copy_get
      :invalid
    end
  end.new.freeze

  ce = PG::TextDecoder::CopyRow.new(type_map: tm).freeze
  expect { ce.decode("5\t6\n") }.to raise_error(TypeError, /kind of Integer/)
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_map_in_ruby_spec.rb

Классы в модуле 1/3

```
before do
  configuration.relation(:users)
  configuration.relation(:tasks)

  class Test::CreateUser < ROM::Commands::Create[:memory]
    config.component.id = :create
    config.component.namespace = :users
    config.component.relation = :users
    config.result = :one
  end
end
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/suite/rom/commands/create_spec.rb

Классы в модуле 2/3

```
it 'raises no error when all monads are loaded' do
  expect do
    class Test::MyClass
      include Dry::Monads
    end
  end.not_to raise_error

  expect(Test::MyClass.constants).to include(:Success)
  expect(Test::MyClass.constants).to include(:Failure)
end
```

https://github.com/dry-rb/dry-monads/blob/704c1b/spec/integration/all_monads_inclusion_spec.rb

Классы в модуле 3/3

```
module Test
  def self.remove_constants
    constants.each(&method(:remove_const))
  end
end

config.after do
  gateway.disconnect if respond_to?(:gateway)
    && gateway.respond_to?(:disconnect)
  Test.remove_constants
end
```

https://github.com/rom-rb/rom/blob/7fb82c/spec/spec_helper.rb

Expectation

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```
it "returns a new instance" do
  expect(none).to eql(None())
  expect(none.trace).to include("spec/integration/\
    maybe_spec.rb:9:in `block")
end
```

https://github.com/dry-rb/dry-monads/blob/704c1b/spec/integration/maybe_spec.rb

Несколько проверок 1/3

```
let!(:textdec_int) { PG::TextDecoder::Integer.new(name: 'INT4', oid: 23).freeze }
let!(:textdec_float) { PG::TextDecoder::Float.new(name: 'FLOAT8', oid: 701).freeze }
let!(:textdec_string) { PG::TextDecoder::String.new(name: 'TEXT', oid: 25).freeze }
let!(:textdec_bytea) { PG::TextDecoder::Bytea.new(name: 'BYTEA', oid: 17).freeze }
let!(:binarydec_float) { PG::BinaryDecoder::Float.new(name: 'FLOAT8', oid: 701, format: 1).freeze }
let!(:tm) do
  tm = PG::TypeMapByOid.new
  tm.add_coder textdec_int
  tm.add_coder textdec_float
  tm.add_coder binarydec_float
  tm.add_coder pass_through_type
  tm.freeze
end
it "should retrieve it's conversions" do
  expect(tm.coders).to eq([
                            textdec_int,
                            textdec_float,
                            pass_through_type,
                            binarydec_float
                          ])
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_map_by_oid_spec.rb

Несколько проверок 2/3

```
it 'should check format when deleting coders' do
  expect { tm_writable.rm_coder(2, 123) }.to raise_error(ArgumentError)
  expect { tm_writable.rm_coder(-1, 123) }.to raise_error(ArgumentError)
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_map_by_oid_spec.rb

Несколько проверок 3/3

```
[1, 0].each do |format|
  it "should convert format #{format} timestamps per TimestampUtcToLocal" do
    regi = PG::BasicTypeRegistry.new
    regi.register_type 0, 'timestamp', nil, PG::TextDecoder::TimestampUtcToLocal
    regi.register_type 1, 'timestamp', nil, PG::BinaryDecoder::TimestampUtcToLocal
    @conn.type_map_for_results = PG::BasicTypeMapForResults.new(@conn, registry: regi)
    res = @conn.exec_params("SELECT CAST('2013-07-31 23:58:59+02' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('1913-12-31 23:58:59.1231-03' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('4714-11-24 23:58:59.1231-03 BC' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('294276-12-31 23:58:59.1231-03' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('infinity' AS TIMESTAMP WITHOUT TIME ZONE),
        CAST('-infinity' AS TIMESTAMP WITHOUT TIME ZONE)", [], format)

    expect(res.getvalue(0, 0).iso8601(3)).to eq(Time.utc(2013, 7, 31, 23, 58, 59).getlocal.iso8601(3))
    expect(res.getvalue(0, 1).iso8601(3)).to eq(Time.utc(1913, 12, 31, 23, 58, 59.1231).getlocal.iso8601(3))
    expect(res.getvalue(0, 2).iso8601(3)).to eq(Time.utc(-4713, 11, 24, 23, 58, 59.1231).getlocal.iso8601(3))
    expect(res.getvalue(0, 3).iso8601(3)).to eq(Time.utc(294_276, 12, 31, 23, 58, 59.1231).getlocal.iso8601(3))
    expect(res.getvalue(0, 4)).to eq('infinity')
    expect(res.getvalue(0, 5)).to eq('-infinity')
  end
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/basic_type_map_for_results_spec.rb

```
specify '#(try_)push(_op)' do
  channel = Concurrent::Promises::Channel.new 1

  expect(channel.size).to eq 0
  expect(channel.try_push(:v1)).to be_truthy
  expect(channel.size).to eq 1
  expect(channel.try_push(:v2)).to be_falsey
  expect(channel.size).to eq 1

  channel = Concurrent::Promises::Channel.new 1
  expect(channel.push(:v1)).to eq channel
  expect(channel.size).to eq 1
  thread = in_thread { channel.push :v2 }
  is_sleeping thread
  expect(channel.size).to eq 1
  channel.pop
  expect(channel.size).to eq 1
  expect(thread.value).to eq channel
  channel.pop
  expect(channel.size).to eq 0

  channel = Concurrent::Promises::Channel.new 1
  expect(channel.push(:v1)).to eq channel
  expect(channel.size).to eq 1
  thread = in_thread { channel.push :v2, 0.01 }
  is_sleeping thread
  expect(channel.size).to eq 1
  expect(thread.value).to eq false
  channel.pop
  expect(channel.size).to eq 0
  expect(channel.push(:v3, 0)).to eq true
  expect(channel.size).to eq 1
  thread = in_thread { channel.push :v2, 1 }
  is_sleeping thread
  channel.pop
  expect(channel.size).to eq 1
  expect(thread.value).to eq true

  # ..
end
```

https://github.com/ruby-concurrency/concurrent-ruby/blob/1982b9/spec/concurrent/edge/channel_spec.rb

Очевидный комментарий

```
it 'sets :max_length to DEFAULT_MAX_POOL_SIZE' do
  expect(subject.max_length).to eq(
    described_class::DEFAULT_MAX_POOL_SIZE
  )
end
```

concurrent-ruby/blob/1982b9/spec/concurrent/executor/cached_thread_pool_spec.rb

Магические циклы

```
it 'should decode integers of different lengths from text format' do
  30.times do |zeros|
    expect(textdec_int.decode('1' + '0' * zeros)).to eq(10**zeros)
    expect(textdec_int.decode(zeros == 0 ? '0' : '9' * zeros)).to eq(10**zeros - 1)
    expect(textdec_int.decode('-1' + '0' * zeros)).to eq(-10**zeros)
    expect(textdec_int.decode(zeros == 0 ? '0' : '-' + '9' * zeros)).to eq(-10**zeros + 1)
  end
  66.times do |bits|
    expect(textdec_int.decode((2**bits).to_s)).to eq(2**bits)
    expect(textdec_int.decode((2**bits - 1).to_s)).to eq(2**bits - 1)
    expect(textdec_int.decode((-2**bits).to_s)).to eq(-2**bits)
    expect(textdec_int.decode((-2**bits + 1).to_s)).to eq(-2**bits + 1)
  end
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_spec.rb

Faker

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```
let(:logger_stream) { StringIO.new }

def configure_logger
  Hanami.app.config.logger.stream = logger_stream
end

def logs
  @logs ||= (logger_stream.rewind and logger_stream.read)
end

expect(logs).to match %r{GET 404 \d+(µs|ms) 127.0.0.1 /}
```

https://github.com/hanami/hanami/blob/675b44/spec/integration/logging/exception_logging_spec.rb

TcpGateSwitcher 1/2

```
# This is a transparent TCP proxy for testing blocking behaviour in a time insensitive way.
#
# It works as a gate between the client and the server, which is enabled or disabled by the spec.
# Data transfer can be blocked per API.
# The TCP communication in a C extension can be verified in a (mostly) timing insensitive way.
# If a call does IO but doesn't handle non-blocking state, the test will block and can be caught by an external timeout.
#
#
#   PG.connect                    TcpGateSwitcher
#    port:5444        .-------------------------------------.
#        .--start/stop---------------> T                     |         DB
#  .-----|-----.      |                | /                   |      .------.
#  |    non-   |      |                |/                    |      |Server|
#  |  blocking |      | TCPServer      /           TCPSocket |      | port |
#  |   specs   |------->port 5444-----/   ---------port 5432------->| 5432 |
#  '-----------'      '-------------------------------------'      '------'
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/helpers/tcp_gate_switcher.rb

TcpGateSwitcher 2/2

```
run_with_gate(200) do |conn, gate|
  conn.setnonblocking(true)

  gate.stop # ТУТ
  data = 'x' * 1000 * 1000 * 30
  res = conn.send_query_params('SELECT LENGTH($1)',
    [data])
  expect(res).to be_nil

  res = conn.flush
  expect(res).to be_falsey

  gate.start # И ЗДЕСЬ
  IO.select(
    nil, [conn.socket_io], [conn.socket_io], 10
  ) until conn.flush
  expect(conn.flush).to be_truthy

  res = conn.get_last_result
  expect(res.values).to eq([[data.length.to_s]])
end
```

https://github.com/ged/ruby-pg/blob/2218eb/spec/pg/connection_spec.rb

Helpers

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`expect_next_instance_of` 1/3

```
context 'when all inputs are correct' do
  it 'imports a repository' do
    # ..

    expect_next_instance_of(Project) do |project|
      expect(project).to receive(:after_import)
    end

    import_task
  end
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/tasks/import_rake_spec.rb

`expect_next_instance_of` 2/3

```
def expect_next_instance_of(klass, *new_args, &blk)
  stub_new(expect(klass), nil, false, *new_args, &blk)
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/support/helpers/next_instance_of.rb

`expect_next_instance_of` 3/3

```
def stub_new(target, number, ordered = false, *new_args, &blk)
  receive_new = receive(:new)
  receive_new.ordered if ordered
  receive_new.with(*new_args) if new_args.present?

  if number.is_a?(Range)
    receive_new.at_least(number.begin).times if number.begin
    receive_new.at_most(number.end).times if number.end
  elsif number
    receive_new.exactly(number).times
  end

  target.to receive_new.and_wrap_original do |*original_args, **original_kwargs|
    method, *original_args = original_args
    method.call(*original_args, **original_kwargs).tap(&blk)
  end
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/support/helpers/next_instance_of.rb

`without_env_vars` 1/2

```
with_env_vars 'XDG_CONFIG_HOME' => '~/.custom-config' do
  options = parse_options
  expect(options[:formatters]).to eq([['overridden_xdg']])
end

without_env_vars 'XDG_CONFIG_HOME' do
  options = parse_options
  expect(options[:formatters]).to eq([['default_xdg']])
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/rspec/core/configuration_options_spec.rb

`without_env_vars` 2/2

```
def without_env_vars(*vars)
  original = ENV.to_hash
  vars.each { |k| ENV.delete(k) }

  begin
    yield
  ensure
    ENV.replace(original)
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/spec_helper.rb

Выразительность

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Factory Alias

```
FactoryBot.define do
  factory :user, aliases: [:author, :assignee, :recipient,
                           :owner, :resource_owner] do
    email { generate(:email) }
    name { generate(:name) }
    # ..
  end
end
```

https://github.com/gitlabhq/gitlabhq/blob/652dfd/spec/factories/users.rb

Метод с ожиданием 1/2

```
def textdec_timestamptz_decode_should_fail(str)
  expect(textdec_timestamptz.decode(str)).to eq(str)
end

it 'fails when the timestamp is an empty string' do
  textdec_timestamptz_decode_should_fail('')
end
```

https://github.com/ged/ruby-pg/blob/1c67bb/spec/pg/type_spec.rb

Метод с ожиданием 2/2

```
def expect_parsing_to_fail_mentioning_source(source, options = [])
  expect do
    parse_options(*options)
  end.to raise_error(SystemExit).and output(a_string_including(
                                              "invalid option: --foo_bar (defined in #{source})",
                                              'Please use --help for a listing of valid options'
                                            )).to_stderr
end

context 'defined in $XDG_CONFIG_HOME/rspec/options' do
  it 'mentions the file name in the error so users know where to look for it' do
    file_name = File.expand_path('~/.config/rspec/options')
    create_fixture_file(file_name, '--foo_bar')
    expect_parsing_to_fail_mentioning_source(file_name)
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/rspec/core/configuration_options_spec.rb

Детали реализации

```
[
  ['--failure-exit-code', '3', :failure_exit_code, 3],
  ['--pattern', 'foo/bar', :pattern, 'foo/bar'],
  ['--failure-exit-code', '37', :failure_exit_code, 37],
  ['--default-path', 'behavior', :default_path, 'behavior'],
  ['--order', 'rand', :order, 'rand'],
  ['--seed', '37', :order, 'rand:37'],
  ['--drb-port', '37', :drb_port, 37]
].each do |cli_option, cli_value, config_key, config_value|
  it "forces #{config_key}" do
    opts = config_options_object(cli_option, cli_value)
    expect(config).to receive(:force) do |pair|
      expect(pair.keys.first).to eq(config_key)
      expect(pair.values.first).to eq(config_value)
    end
    opts.configure(config)
  end
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/rspec/core/configuration_options_spec.rb

rspec-parameterized 1/3

```
where(:dot_com, :request_user_agent, :qa_user_agent, :result) do
  false | 'qa_user_agent' | 'qa_user_agent' | false
  true  | nil             | 'qa_user_agent' | false
  true  | ''              | 'qa_user_agent' | false
  true  | 'qa_user_agent' | ''              | false
  true  | 'qa_user_agent' | nil             | false
  true  | 'qa_user_agent' | 'qa_user_agent' | true
end
```

https://github.com/gitlabhq/gitlabhq/blob/d1ad38/spec/lib/gitlab/qa_spec.rb

rspec-parameterized 2/3

```
with_them do
  before do
    allow(Gitlab).to receive(:com?).and_return(dot_com)
    stub_env('GITLAB_QA_USER_AGENT', qa_user_agent)
  end

  subject { described_class.request?(request_user_agent) }

  it { is_expected.to eq(result) }
end
```

https://github.com/gitlabhq/gitlabhq/blob/d1ad38/spec/lib/gitlab/qa_spec.rb

rspec-parameterized 3/3

```
describe "Hash arguments" do
  where(a: [1, 3], b: [5, 7, 9], c: [2, 4])

  with_them do
    it "sums is even" do
      expect(a + b + c).to be_even
    end
  end
end
```

tomykaira/rspec-parameterized

Acceptance

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```ruby
specify 'types from a provided types module can be used as setting constructors to coerce values' do
  with_tmp_directory(Dir.mktmpdir) do
    write 'config/app.rb', <<~RUBY
      require "hanami"

      module TestApp
        class App < Hanami::App
        end
      end
    RUBY

    write 'config/settings.rb', <<~RUBY
      module TestApp
        class Settings < Hanami::Settings
          Bool = Types::Params::Bool

          setting :numeric, constructor: Types::Params::Integer
          setting :flag, constructor: Bool
        end
      end
    RUBY

    ENV['NUMERIC'] = '42'
    ENV['FLAG'] = 'true'

    require 'hanami/prepare'

    expect(Hanami.app['settings'].numeric).to eq 42
    expect(Hanami.app['settings'].flag).to be true
  end
end
```

https://github.com/hanami/hanami/blob/a2bdb7/spec/integration/settings/using_types_spec.rb

```
context 'when used in a multithreaded environment with a cassette',
                                        with_monkey_patches: :excon do
  it 'properly stubs threaded requests' do
    VCR.use_cassette('/foo') do
      threads = 50.times.map do
        Thread.start do
          Excon.get "http://localhost:#{VCR::SinatraApp.port}/foo"
        end
      end
      threads.each(&:join)
    end

    expect(
      recorded_content_for('foo')
    ).to include('FOO!')
  end
end
```

https://github.com/vcr/vcr/blob/8f5636/spec/acceptance/concurrency_spec.rb

```ruby
def passing_example(fail_if_no_examples)
  "
    RSpec.configure { |c| c.fail_if_no_examples = #{fail_if_no_examples} }

    RSpec.describe 'something' do
      it 'succeeds' do
        true
      end
    end
  "
end

it 'succeeds if fail_if_no_examples set to true' do
  write_file 'spec/example_spec.rb', passing_example(true)
  run_command ''
  expect(last_cmd_stdout).to include('1 example, 0 failures')
  expect(last_cmd_exit_status).to eq(0)
end
```

https://github.com/rspec/rspec-core/blob/1eeadc/spec/integration/fail_if_no_examples_spec.rb

Итоги

1. тестируй
2. тестируй важное
3. читай чужой код

Сылки

1. https://bit.ly/3QSGIxh "RSpec. Поддерживаемость"
2. https://github.com/jeremyevans/ruby-warning
3. https://github.com/test-prof/test-prof
4. https://github.com/rspec-parameterized/rspec-parameterized-core

Спасибо!